Analysis of the Association Between Variations in Obesity Prevalence Among Year 6 Children Across London Boroughs and Socioeconomic Factors

Preparation

- Github link *[Optional]*
- Number of words: 1444
- Runtime: 25 seconds (*Memory 32 GB, CPU AMD Ryzen AI 7 H 350 w/ Radeon 860M (2.00 GHz)*)
- Coding environment: Podman (JupyterLab)
- License: this notebook is made available under the Creative Commons Attribution license (or other license that you like).
- Additional library *[libraries not included in SDS Docker or not used in this module]*:
  - **watermark**: A Jupyter Notebook extension for printing timestamps, version numbers, and hardware information.
  - .....

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Introduction

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Childhood obesity is an ongoing public health concern in England, as it not only affects children's current and future health, but also reflects wider social and spatial inequalities. The National Child Measurement Programme has been monitoring children's weight for a long time. The latest official statistics show that the obesity rate among Year 6 children is relatively high in England, indicating that research on this issue at the local level is still of practical significance (NHS, 2024).

Some studies have generally pointed to a stable association between childhood obesity and socio-economic disadvantage. Kinra, Nelder, and Lewendon (2000) found in a UK urban study that children in poorer areas were more likely to be obese. Conrad and Capewell (2012) further show that there is a significant positive correlation between childhood overweight and obesity rates and area deprivation on a local scale in England. White, Rehkopf and Mortensen (2016) found that socioeconomic inequalities in childhood obesity did not decrease over a period. Broadbent et al., (2024) further suggest that this inequality is persistent. It can be seen that childhood obesity is not just an individual behavioral problem, but is also closely related to the local socio-economic environment.

Based on this, this essay uses the London ward as the unit of analysis to study differences in the prevalence of obesity among Year 6 students in different administrative districts, and focuses on examining the relationship between several local socioeconomic variables and their changes.

Research questions

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Which ward-level socio-economic factors are most strongly associated with differences in the prevalence of obesity among Year 6 students in London?

Data

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This study uses the Greater London Authority (GLA) Ward Profiles dataset as the primary data source.During the data cleaning process, only observations at the ward level were retained, whilst summary rows and records with missing key variables were deleted; the final sample comprised 612 wards.The dependent variable was the prevalence of obesity among sixth-grade children, and the median household income, employment rate, proportion of the population without formal education, and proportion of dependent children in unemployed households were selected as explanatory variables.

In [1]:
```
# ========================
# Load, clean, prepare the data
# ========================
import pandas as pd
```

```python
import numpy as np

# 1. Read raw data
url = "https://raw.githubusercontent.com/CHENYIZHAO1203/CASA006/main/Data/ward-p
df = pd.read_csv(url, encoding="latin1", low_memory=False)
# if read data local
## df = pd.read_csv("Data/ward-profiles-excel-version.csv", encoding="latin1", l

# 2. Retain only ward level (New code starts with E05)
df_ward = df[df["New code"].astype(str).str.startswith("E05")].copy()

# 3. Select the variables
cols = [
    "Ward name",
    "New code",
    "% children in year 6 who are obese- 2011/12 to 2013/14",
    "Median Household income estimate (2012/13)",
    "Employment rate (16-64) - 2011",
    "% with no qualifications - 2011",
    "% dependent children (0-18) in out-of-work households - 2014",
    "Claimant rate of key out-of-work benefits (working age client group) (2014)
    "Claimant Rate of Housing Benefit (2015)",
    "% of lone parents not in employment - 2011",
    "Rate of JobSeekers Allowance (JSA) Claimants - 2015"
]

df_ward = df_ward[cols].copy()

# 4. Rename the variable to facilitate subsequent analysis
df_ward = df_ward.rename(columns={
    "Ward name": "ward_name",
    "New code": "ward_code",
    "% children in year 6 who are obese- 2011/12 to 2013/14": "year6_obesity_pct
    "Median Household income estimate (2012/13)": "median_income",
    "Employment rate (16-64) - 2011": "employment_rate",
    "% with no qualifications - 2011": "no_qual_pct",
    "% dependent children (0-18) in out-of-work households - 2014": "child_workl
    "Claimant rate of key out-of-work benefits (working age client group) (2014)
    "Claimant Rate of Housing Benefit (2015)": "housing_benefit_rate",
    "% of lone parents not in employment - 2011": "loneparent_noemp_pct",
    "Rate of JobSeekers Allowance (JSA) Claimants - 2015": "jsa_rate"
})

# 5. Remove observations with missing values for the dependent variable
df_ward = df_ward.dropna(subset=["year6_obesity_pct"]).copy()

# 6. Review of basic data
print("Dataset shape:", df_ward.shape)
print("\nMissing values:")
print(df_ward.isna().sum())
```

```
Dataset shape: (612, 11)

Missing values:
ward_name                 0
ward_code                 0
year6_obesity_pct         0
median_income             0
employment_rate           0
no_qual_pct               0
child_workless_pct        0
outofwork_benefit_rate    0
housing_benefit_rate      0
loneparent_noemp_pct      0
jsa_rate                  0
dtype: int64
```

In [2]:
```python
# Look at the previous lines to check the cleaned data
display(df_ward.head())
```

	ward_name	ward_code	year6_obesity_pct	median_income	employment_rate	no
1	Barking and Dagenham - Abbey	E05000026	24.7	33920.0	60.3	
2	Barking and Dagenham - Alibon	E05000027	26.0	32470.0	63.1	
3	Barking and Dagenham - Becontree	E05000028	29.3	33000.0	61.2	
4	Barking and Dagenham - Chadwell Heath	E05000029	24.6	33920.0	63.6	
5	Barking and Dagenham - Eastbrook	E05000030	21.6	37400.0	68.2	

In addition, for the purposes of spatial visualisation, this paper utilises boundary data for London wards to create a choropleth map illustrating the spatial distribution of obesity prevalence among Year 6 pupils.

In [3]:
```python
# ========================
# Read ward shapefile
# ========================
import geopandas as gpd
import matplotlib.pyplot as plt

# 1. Read the ward shapefile
wards_gdf = gpd.read_file(
    "https://raw.githubusercontent.com/CHENYIZHAO1203/CASA006/main/Data/London_W
)
# if read data local
## wards_gdf = gpd.read_file("Data/statistical-gis-boundaries-london/ESRI/London
```

```python
# 2. Checking column names
print(wards_gdf.columns.tolist())
display(wards_gdf.head())

# 3. Rename join key
wards_gdf = wards_gdf.rename(columns={"GSS_CODE": "ward_code"})

# 4. Merging Attribute Data
map_gdf = wards_gdf.merge(
    df_ward,
    on="ward_code",
    how="left"
)

print("Map dataframe shape:", map_gdf.shape)
print("Missing obesity values after join:", map_gdf["year6_obesity_pct"].isna().
```

```
['NAME', 'GSS_CODE', 'HECTARES', 'NONLD_AREA', 'LB_GSS_CD', 'BOROUGH', 'POLY_ID',
'geometry']
```

	NAME	GSS_CODE	HECTARES	NONLD_AREA	LB_GSS_CD	BOROUGH	POLY_
0	Chessington South	E05000405	755.173	0.0	E09000021	Kingston upon Thames	508
1	Tolworth and Hook Rise	E05000414	259.464	0.0	E09000021	Kingston upon Thames	1171
2	Berrylands	E05000401	145.390	0.0	E09000021	Kingston upon Thames	504
3	Alexandra	E05000400	268.506	0.0	E09000021	Kingston upon Thames	504
4	Beverley	E05000402	187.821	0.0	E09000021	Kingston upon Thames	1171

```
Map dataframe shape: (625, 18)
Missing obesity values after join: 13
```

The main six variables showed below:

Variable	Type	Description	Notes
year6_obesity_pct	Numeric	Percentage of Year 6 children classified as obese in each ward	
median_income	Numeric	Median household income in each ward	
employment_rate	Numeric	Employment rate of the working-age population	
no_qual_pct	Numeric	Percentage of residents with no formal qualifications	
child_workless_pct	Numeric	Proportion of dependent children living in workless households	
ward_code	Categorical	Unique ward code used to merge attribute data with spatial boundaries	

Methodology

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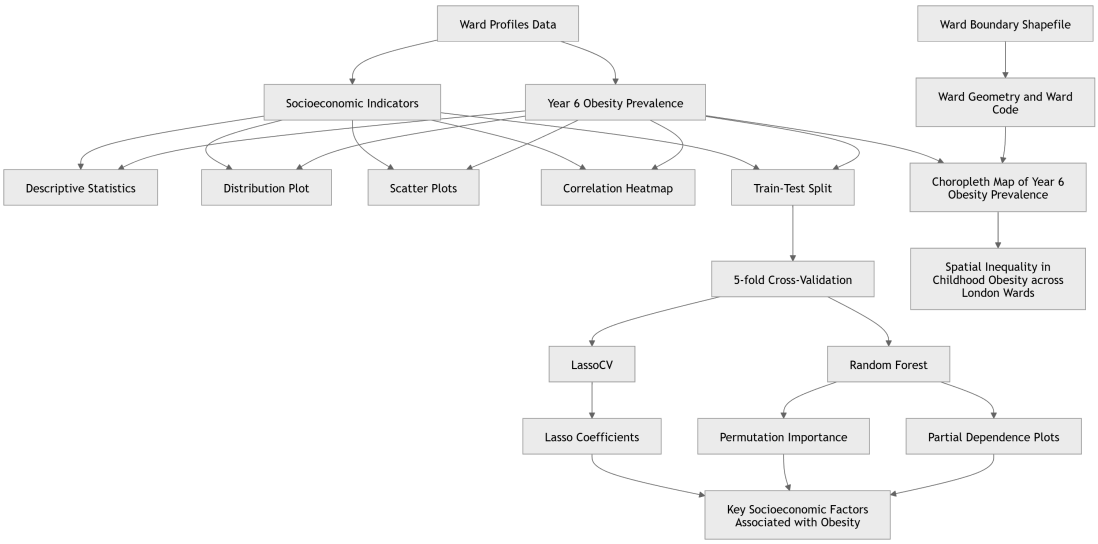
Source: see github link.



This essay uses ward-level cross-sectional ecological analyses. The goal of the study is to identify the socioeconomic characteristics that underlie differences in childhood obesity prevalence between wards, and therefore this paper discusses associations at the area level rather than causality at the individual level.

The analysis consisted of four steps. 1. Using descriptive statistics, distribution plots, scatter plots, correlation heatmaps, and graded color maps, a preliminary description of the prevalence of childhood obesity and its spatial variations was conducted. 2. Lasso regression model is constructed as a sparse linear benchmark for testing the main variables in a linear framework. 3. Construct a Random Forest regression model to capture potential nonlinear relationships and interactions between variables. 4. Interpret the tree model using permutation importance and

partial dependence plots to identify the most important variables and observe the patterns of relationships between key variables and the predicted outcomes.

Model evaluation uses a training and test set split, with 5-fold cross-validation performed on the training set. Model performance is compared using RMSE, MAE, and R^2.

Results and discussion

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In [4]:
```
# Generate the main analysis variable table
main_vars = [
    "year6_obesity_pct",
    "median_income",
    "employment_rate",
    "no_qual_pct",
    "child_workless_pct"
]

desc_table = df_ward[main_vars].describe().T
desc_table = desc_table[["count", "mean", "std", "min", "25%", "50%", "75%", "ma

print("\nTable 1: Descriptive statistics for main variables:")
display(desc_table)
```

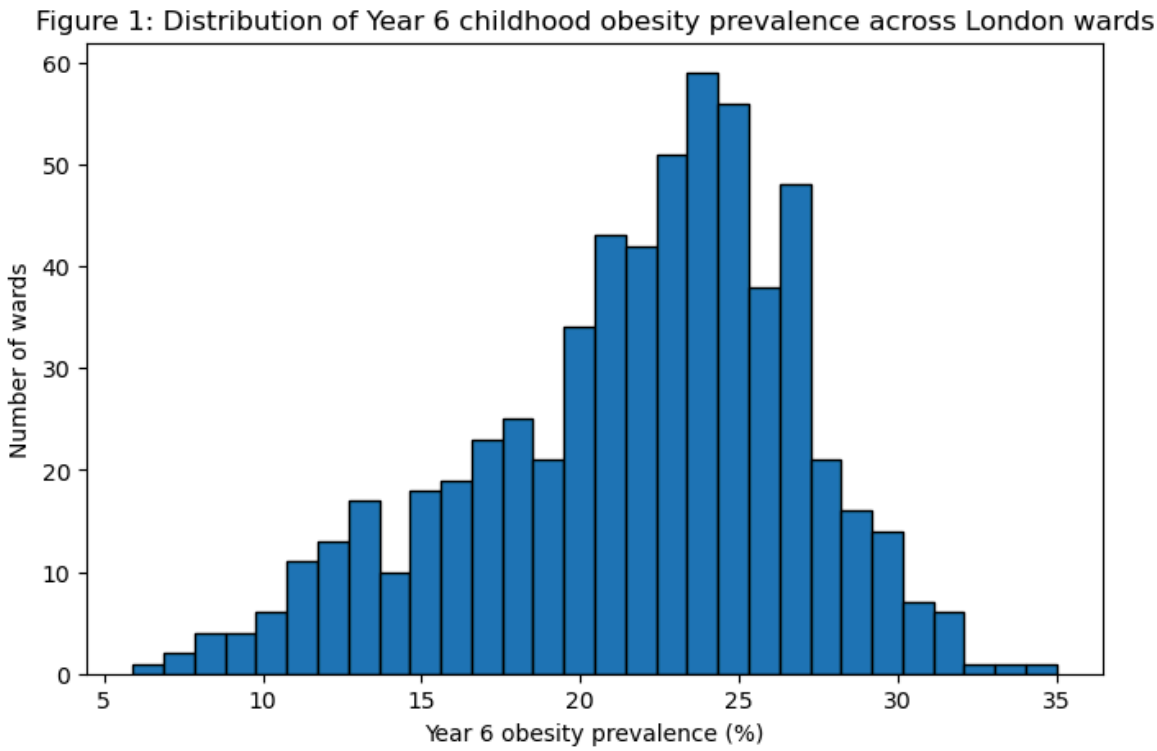
```
Table 1: Descriptive statistics for main variables:
```

	count	mean	std	min	25%	50%	75%
year6_obesity_pct	612.0	21.75	5.15	5.9	18.58	22.7	25.32
median_income	612.0	38781.70	6634.95	25090.0	33562.50	38080.0	43130.00
employment_rate	612.0	69.45	6.48	46.1	64.90	70.2	74.30
no_qual_pct	612.0	17.87	5.86	4.5	13.78	17.6	21.62
child_workless_pct	612.0	15.91	7.37	2.2	9.70	15.9	21.52

Table 1 shows that the prevalence of obesity among Year 6 children across 612 London wards averaged 21.75%, ranging from 5.9% to 35.0%, indicating significant variation between wards.The core socioeconomic variables also exhibit strong spatial heterogeneity, for example, median household income ranges from £25,090 to £65,500, and the proportion of dependent children in unemployed households ranges from 2.2% to 35.5%.

In [5]:
```
# ========================
# Distribution of the outcome + bivariate scatter plots
# ========================

import matplotlib.pyplot as plt

# 1. Distribution plot of the dependent variable
```

```
plt.figure(figsize=(8, 5))
plt.hist(df_ward["year6_obesity_pct"].dropna(), bins=30, edgecolor="black")
plt.xlabel("Year 6 obesity prevalence (%)")
plt.ylabel("Number of wards")
plt.title("Figure 1: Distribution of Year 6 childhood obesity prevalence across
plt.show()

# 2. Scatter plot of the four core independent variables and the dependent varia
core_predictors = [
    "median_income",
    "employment_rate",
    "no_qual_pct",
    "child_workless_pct"
]

## for var in core_predictors:
for i, var in enumerate(core_predictors, 2):
    temp = df_ward[[var, "year6_obesity_pct"]].dropna()

    plt.figure(figsize=(7, 5))
    plt.scatter(temp[var], temp["year6_obesity_pct"], alpha=0.7)
    plt.xlabel(var)
    plt.ylabel("Year 6 obesity prevalence (%)")
    plt.title(f"Figure {i}: Year 6 obesity prevalence vs {var}")
    plt.show()

# 3. Table of Correlation Coefficients
corr_table = df_ward[["year6_obesity_pct"] + core_predictors].corr()[["year6_obe
corr_table = corr_table.round(3)

print("Table 2: Correlation with Year 6 obesity prevalence:")
display(corr_table)
```



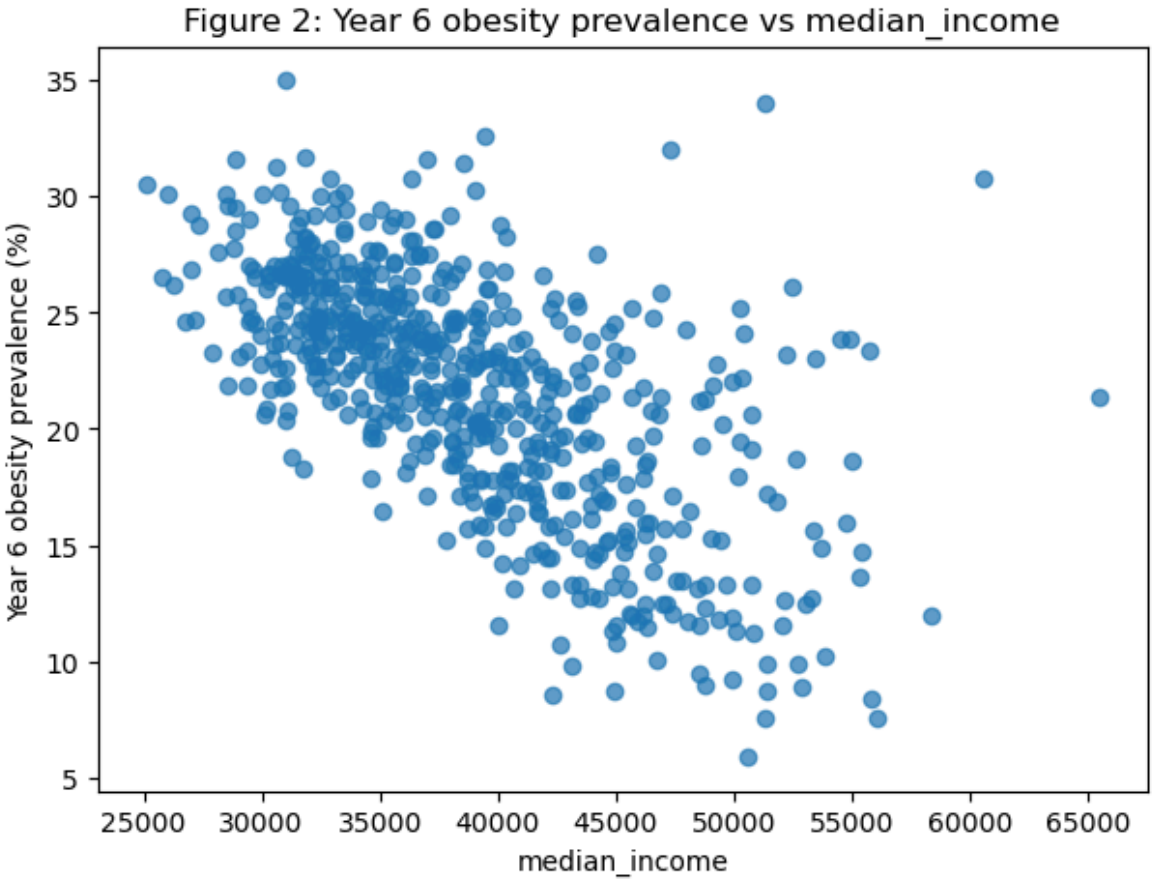


Figure 2: Year 6 obesity prevalence vs median_income
Year 6 obesity prevalence (%)
median_income
35
30
25
20
15
10
5
25000
30000
35000
40000
45000
50000
55000
60000
65000

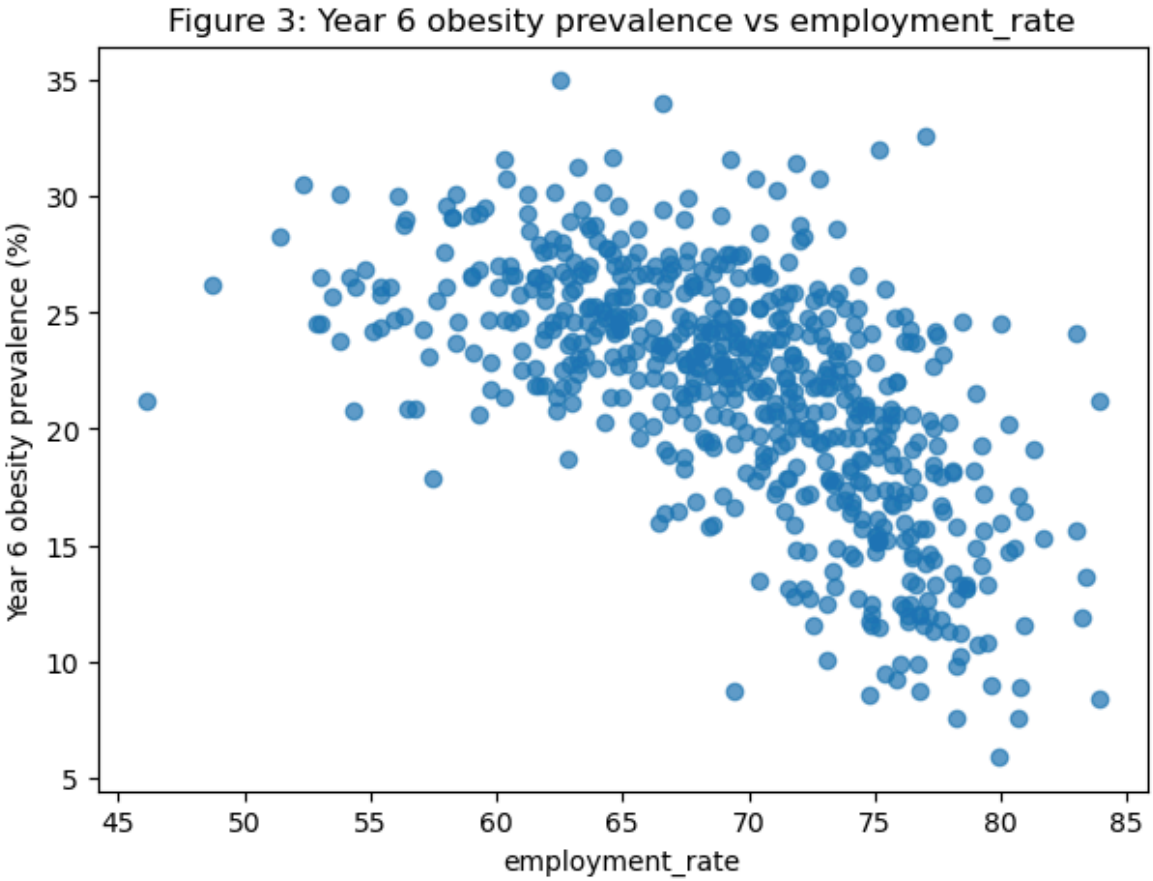
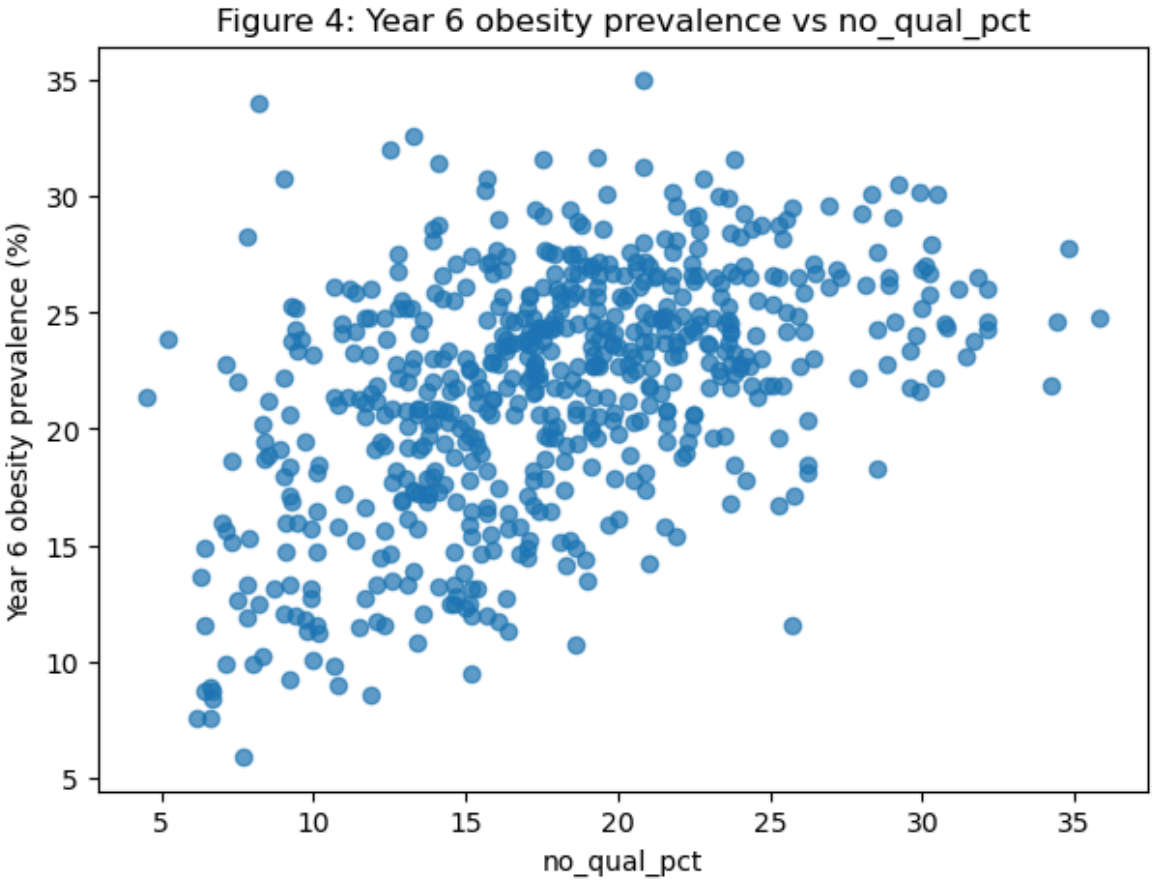
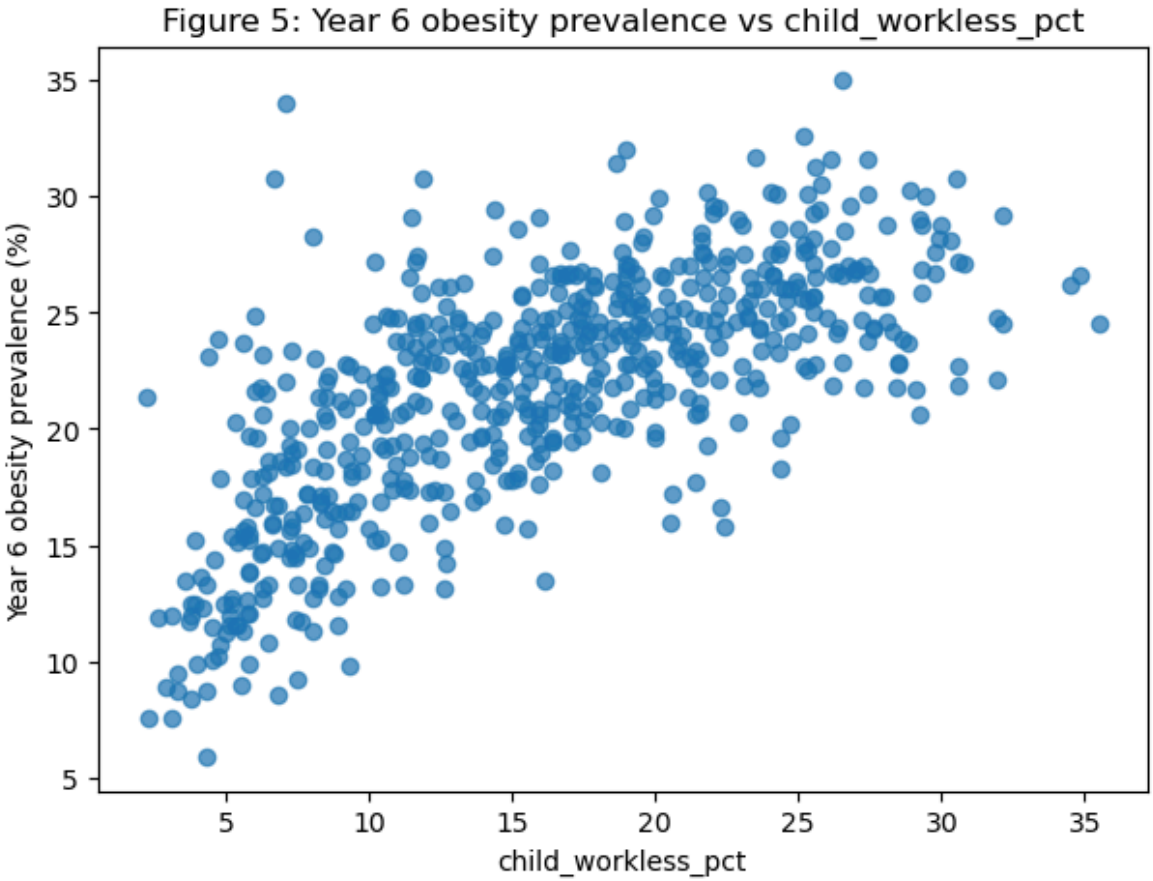


Figure 3: Year 6 obesity prevalence vs employment_rate
Year 6 obesity prevalence (%)
employment_rate
35
30
25
20
15
10
5
45
50
55
60
65
70
75
80
85






```
Table 2: Correlation with Year 6 obesity prevalence:
```

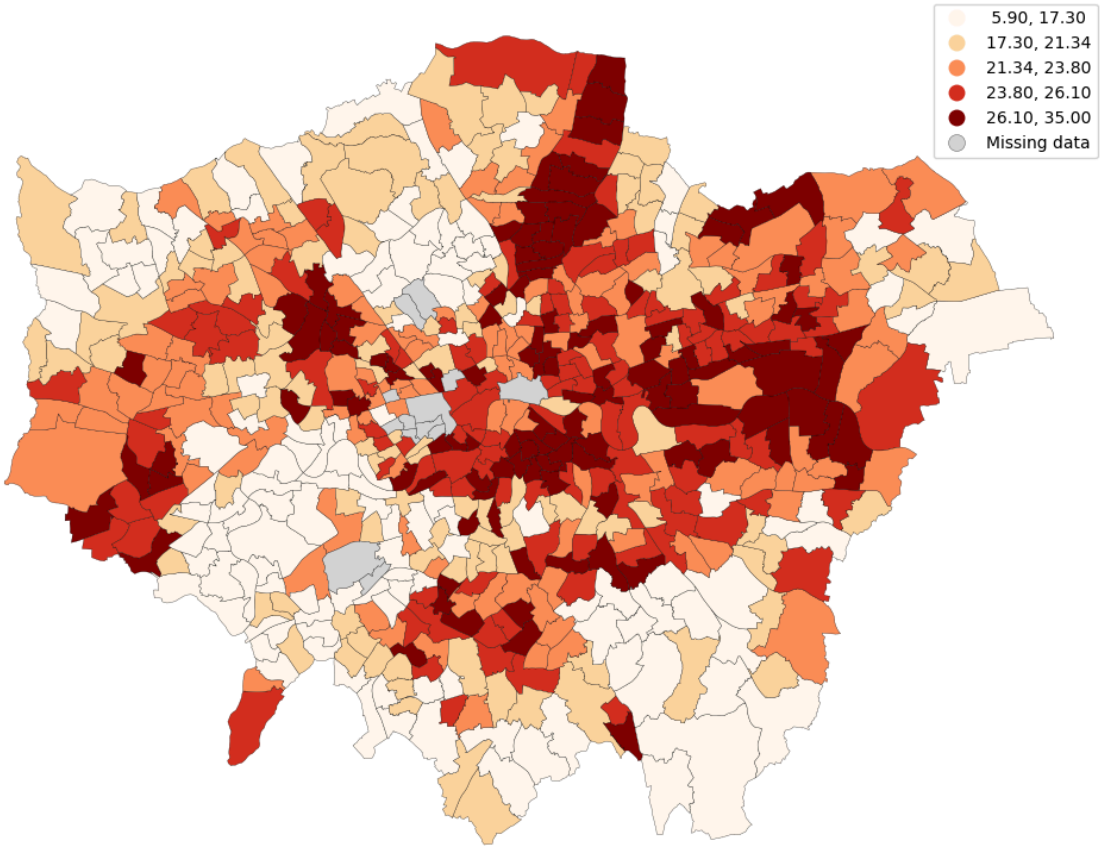
	year6_obesity_pct
year6_obesity_pct	1.000
median_income	-0.637
employment_rate	-0.618
no_qual_pct	0.478
child_workless_pct	0.700

Figure 1 shows that the prevalence of obesity among children in Year 6 is predominantly centred around 20 per cent, although there are a small number of very low and high wards at either end of the scale, suggesting that the indicator is not homogenous within London.

The scatter plots in Figures 2 to 5 further show that the prevalence of childhood obesity is negatively correlated with median household income and employment rate, with correlation coefficients of -0.637 and -0.618, respectively; and positively correlated with the proportion of the population without formal education and the proportion of dependent children in unemployed households, with correlation coefficients of 0.478 and 0.700, respectively. Among these, child_workless_pct has the strongest correlation, suggesting that household work poverty may be the most critical explanatory factor.

In [6]:
```
# ========================
#  choropleth map
# ========================
fig, ax = plt.subplots(figsize=(10, 10))

map_gdf.plot(
    column="year6_obesity_pct",
    scheme="Quantiles",
    k=5,
    cmap="OrRd",
    linewidth=0.2,
    edgecolor="black",
    legend=True,
    missing_kwds={
        "color": "lightgrey",
        "label": "Missing data"
    },
    ax=ax
)

ax.set_title("Figure 6: Year 6 childhood obesity prevalence across London wards"
ax.axis("off")

plt.tight_layout()
plt.show()
```

Figure 6: Year 6 childhood obesity prevalence across London wards

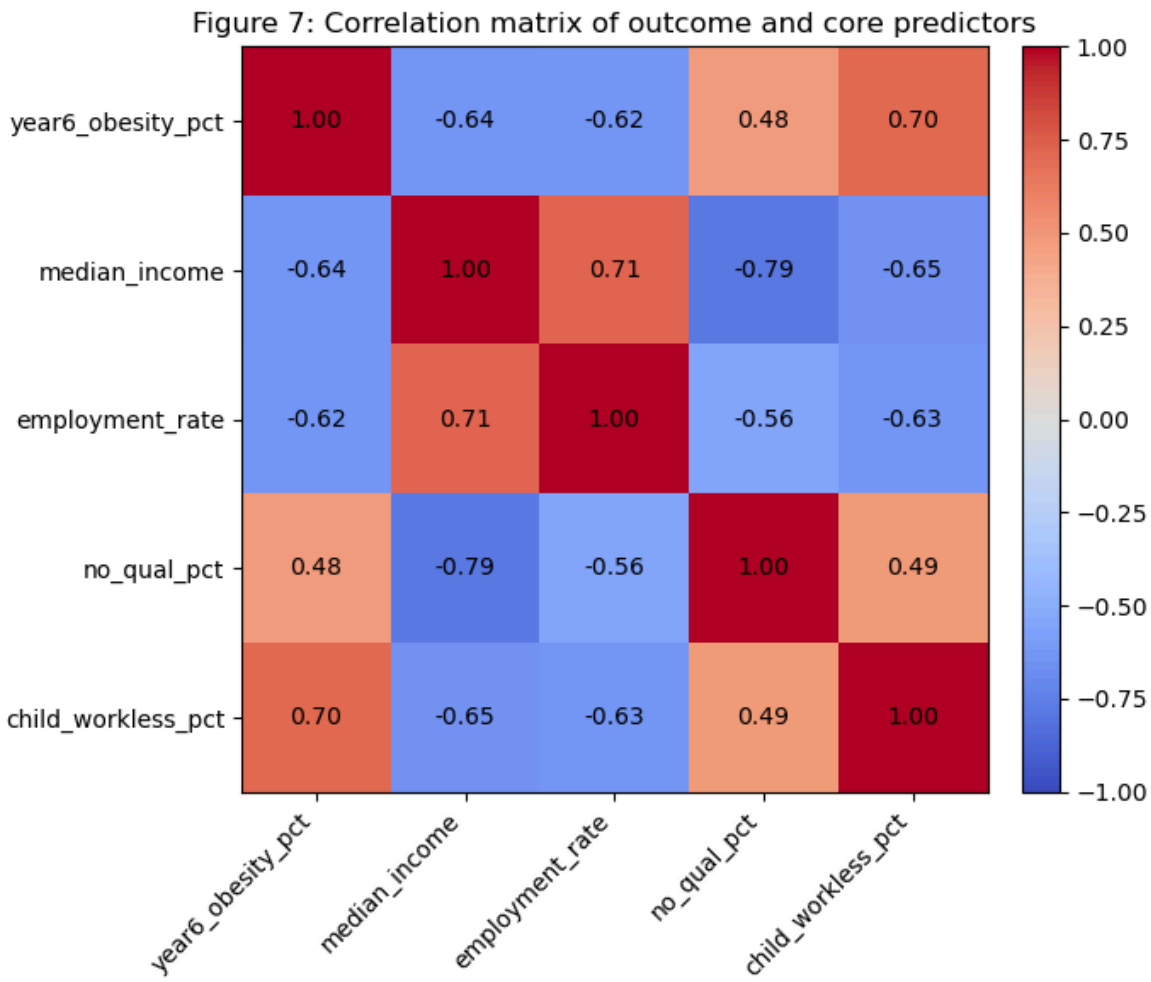


The choropleth map in Figure 6 further shows that the high values of childhood obesity ward are not randomly distributed, but are concentrated in areas with relatively weak socio-economic conditions, showing obvious spatial clustering characteristics. This result is largely consistent with the previous statistical analysis and further illustrates the close link between childhood obesity and socioeconomic inequality within London.

In [7]:
```
# ========================
# Correlation heatmap + train/test split + LassoCV
# ========================

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LassoCV
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

# 1. Correlation heatmap
heatmap_vars = [
    "year6_obesity_pct",
    "median_income",
    "employment_rate",
    "no_qual_pct",
    "child_workless_pct"
```

```
]

corr_matrix = df_ward[heatmap_vars].corr()

plt.figure(figsize=(7, 6))
im = plt.imshow(corr_matrix, cmap="coolwarm", vmin=-1, vmax=1)

plt.xticks(range(len(heatmap_vars)), heatmap_vars, rotation=45, ha="right")
plt.yticks(range(len(heatmap_vars)), heatmap_vars)
plt.title("Figure 7: Correlation matrix of outcome and core predictors")

# Write the correlation coefficient in the grid
for i in range(len(heatmap_vars)):
    for j in range(len(heatmap_vars)):
        plt.text(
            j, i,
            f"{corr_matrix.iloc[i, j]:.2f}",
            ha="center", va="center", fontsize=10
        )

plt.colorbar(im, fraction=0.046, pad=0.04)
plt.tight_layout()
plt.show()

# 2. Preparing modeling data
predictors = [
    "median_income",
    "employment_rate",
    "no_qual_pct",
    "child_workless_pct"
]

X = df_ward[predictors].copy()
y = df_ward["year6_obesity_pct"].copy()

# 3. Divide the training set and test set
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("X_train shape:", X_train.shape)
print("X_test shape:", X_test.shape)

# 4. Create LassoCV pipeline
lasso_model = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler()),
    ("lasso", LassoCV(cv=5, random_state=42, max_iter=10000))
])

# 5. Train Models
lasso_model.fit(X_train, y_train)

# 6. Anticipate
y_train_pred = lasso_model.predict(X_train)
y_test_pred = lasso_model.predict(X_test)

# 7. Assessment of indicators
lasso_train_rmse = np.sqrt(mean_squared_error(y_train, y_train_pred))
lasso_test_rmse = np.sqrt(mean_squared_error(y_test, y_test_pred))
```

```python
lasso_train_mae = mean_absolute_error(y_train, y_train_pred)
lasso_test_mae = mean_absolute_error(y_test, y_test_pred)

lasso_train_r2 = r2_score(y_train, y_train_pred)
lasso_test_r2 = r2_score(y_test, y_test_pred)

lasso_results = pd.DataFrame({
    "Metric": ["RMSE", "MAE", "R2"],
    "Train": [lasso_train_rmse, lasso_train_mae, lasso_train_r2],
    "Test": [lasso_test_rmse, lasso_test_mae, lasso_test_r2]
}).round(3)

print("\nTable 3: LassoCV performance:")
display(lasso_results)

# 8. Extract factor
lasso_estimator = lasso_model.named_steps["lasso"]

lasso_coef = pd.DataFrame({
    "variable": predictors,
    "coefficient": lasso_estimator.coef_
}).sort_values("coefficient", key=np.abs, ascending=False)

print("Best alpha selected by LassoCV:", round(lasso_estimator.alpha_, 5))
print("\nTable 4: Lasso coefficients:")
display(lasso_coef.round(4))

# 9. Coefficient visualisation
plt.figure(figsize=(7, 5))
plt.barh(lasso_coef["variable"], lasso_coef["coefficient"])
plt.xlabel("Coefficient")
plt.ylabel("Variable")
plt.title("Figure 8: LassoCV coefficients")
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```




```
X_train shape: (489, 4)
X_test shape: (123, 4)
```

```
Table 3: LassoCV performance:
```

	Metric	Train	Test
0	RMSE	3.437	3.329
1	MAE	2.689	2.547
2	R2	0.530	0.647

```
Best alpha selected by LassoCV: 0.0735
```

```
Table 4: Lasso coefficients:
```

	variable	coefficient
3	child_workless_pct	2.1508
0	median_income	-0.9876
1	employment_rate	-0.8850
2	no_qual_pct	-0.0000

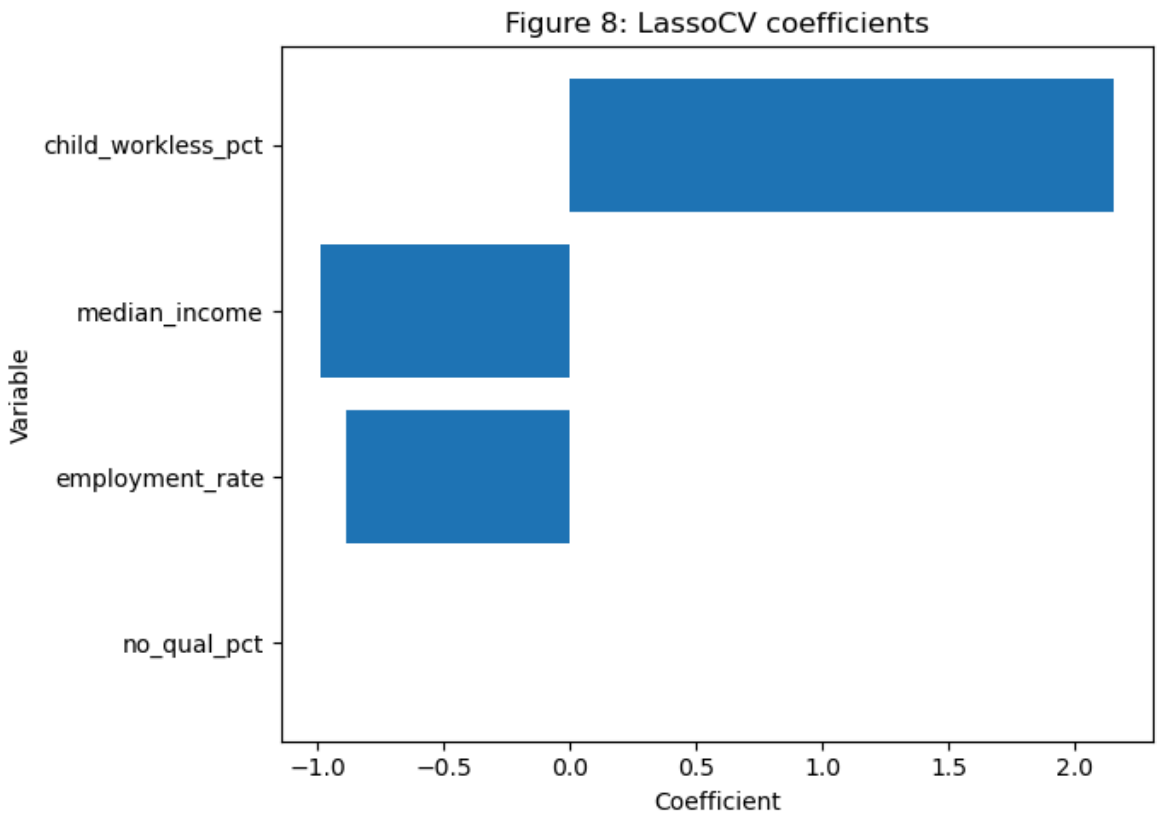


Figure 6 is consistent with these results and also suggests that there is some overlap between some of the explanatory variables.

Table 3 shows that LassoCV has an RMSE of 3.329, a MAE of 2.547, and an R^2 of 0.647 on the test set.Figure 8 shows that child_workless_pct has the most pronounced positive effect, median_income and employment_rate are negative, and no_qual_pct is compressed to 0, indicating a weak independent linear contribution.

In [8]:
```
# ========================
# Random Forest regressor + model comparison
# ========================

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

# 1. Create Random Forest pipeline
rf_pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("rf", RandomForestRegressor(random_state=42))
])

# 2. Setting the parameter grid
param_grid = {
    "rf__n_estimators": [200, 500],
```

```
    "rf__max_depth": [None, 5, 10, 15],
    "rf__min_samples_split": [2, 5, 10],
    "rf__min_samples_leaf": [1, 2, 4]
}

# 3. Tuning with 5-fold CV
rf_grid = GridSearchCV(
    estimator=rf_pipeline,
    param_grid=param_grid,
    cv=5,
    scoring="neg_root_mean_squared_error",
    n_jobs=-1
)

rf_grid.fit(X_train, y_train)

print("Best RF parameters:")
print(rf_grid.best_params_)

# 4. Fetch the best model
rf_best = rf_grid.best_estimator_

# 5. Anticipate
y_train_pred_rf = rf_best.predict(X_train)
y_test_pred_rf = rf_best.predict(X_test)

# 6. Assessment of indicators
rf_train_rmse = np.sqrt(mean_squared_error(y_train, y_train_pred_rf))
rf_test_rmse = np.sqrt(mean_squared_error(y_test, y_test_pred_rf))

rf_train_mae = mean_absolute_error(y_train, y_train_pred_rf)
rf_test_mae = mean_absolute_error(y_test, y_test_pred_rf)

rf_train_r2 = r2_score(y_train, y_train_pred_rf)
rf_test_r2 = r2_score(y_test, y_test_pred_rf)

rf_results = pd.DataFrame({
    "Metric": ["RMSE", "MAE", "R2"],
    "Train": [rf_train_rmse, rf_train_mae, rf_train_r2],
    "Test": [rf_test_rmse, rf_test_mae, rf_test_r2]
}).round(3)

print("\nTable 5: Random Forest performance:")
display(rf_results)

# 7. Model Comparison Table
comparison_table = pd.DataFrame({
    "Model": ["LassoCV", "Random Forest"],
    "Test_RMSE": [lasso_test_rmse, rf_test_rmse],
    "Test_MAE": [lasso_test_mae, rf_test_mae],
    "Test_R2": [lasso_test_r2, rf_test_r2]
}).round(3)

print("\nTable 6: Model comparison on the test set:")
display(comparison_table)

# 8. Predicted value vs. true value plots (Lasso)
plt.figure(figsize=(6, 6))
plt.scatter(y_test, y_test_pred, alpha=0.7)
plt.xlabel("Observed obesity prevalence (%)")
```

```
plt.ylabel("Predicted obesity prevalence (%)")
plt.title("Figure 9: LassoCV: observed vs predicted")
min_val = min(y_test.min(), y_test_pred.min())
max_val = max(y_test.max(), y_test_pred.max())
plt.plot([min_val, max_val], [min_val, max_val], linestyle="--")
plt.tight_layout()
plt.show()

# 9. Predicted value vs. true value plots (Lasso)
plt.figure(figsize=(6, 6))
plt.scatter(y_test, y_test_pred_rf, alpha=0.7)
plt.xlabel("Observed obesity prevalence (%)")
plt.ylabel("Predicted obesity prevalence (%)")
plt.title("Figure 10: Random Forest: observed vs predicted")
min_val = min(y_test.min(), y_test_pred_rf.min())
max_val = max(y_test.max(), y_test_pred_rf.max())
plt.plot([min_val, max_val], [min_val, max_val], linestyle="--")
plt.tight_layout()
plt.show()
```

```
Best RF parameters:
{'rf__max_depth': 5, 'rf__min_samples_leaf': 1, 'rf__min_samples_split': 10, 'rf_
_n_estimators': 500}
```

Table 5: Random Forest performance:

	Metric	Train	Test
0	RMSE	2.480	3.036
1	MAE	1.979	2.294
2	R2	0.755	0.707

Table 6: Model comparison on the test set:

	Model	Test_RMSE	Test_MAE	Test_R2
0	LassoCV	3.329	2.547	0.647
1	Random Forest	3.036	2.294	0.707

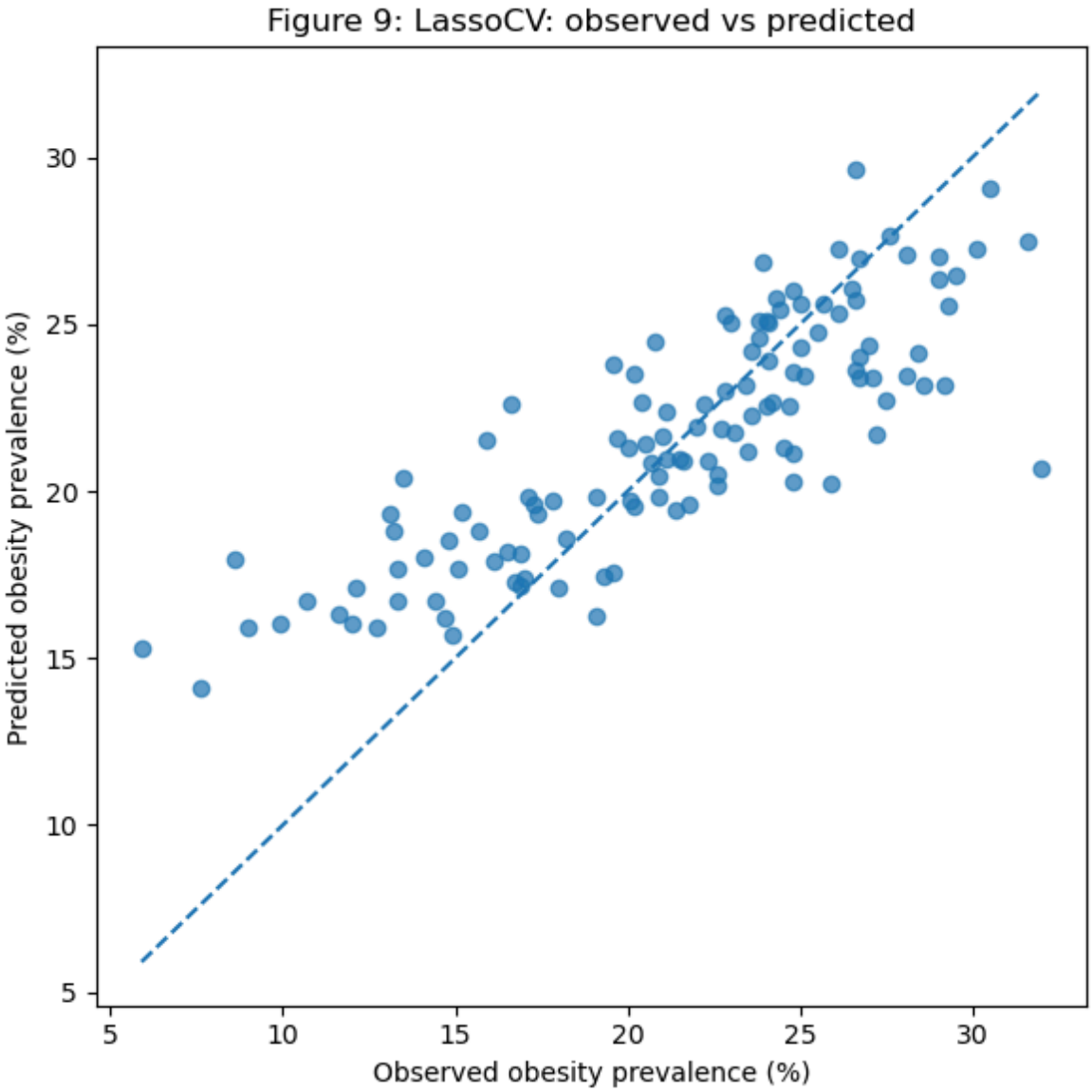
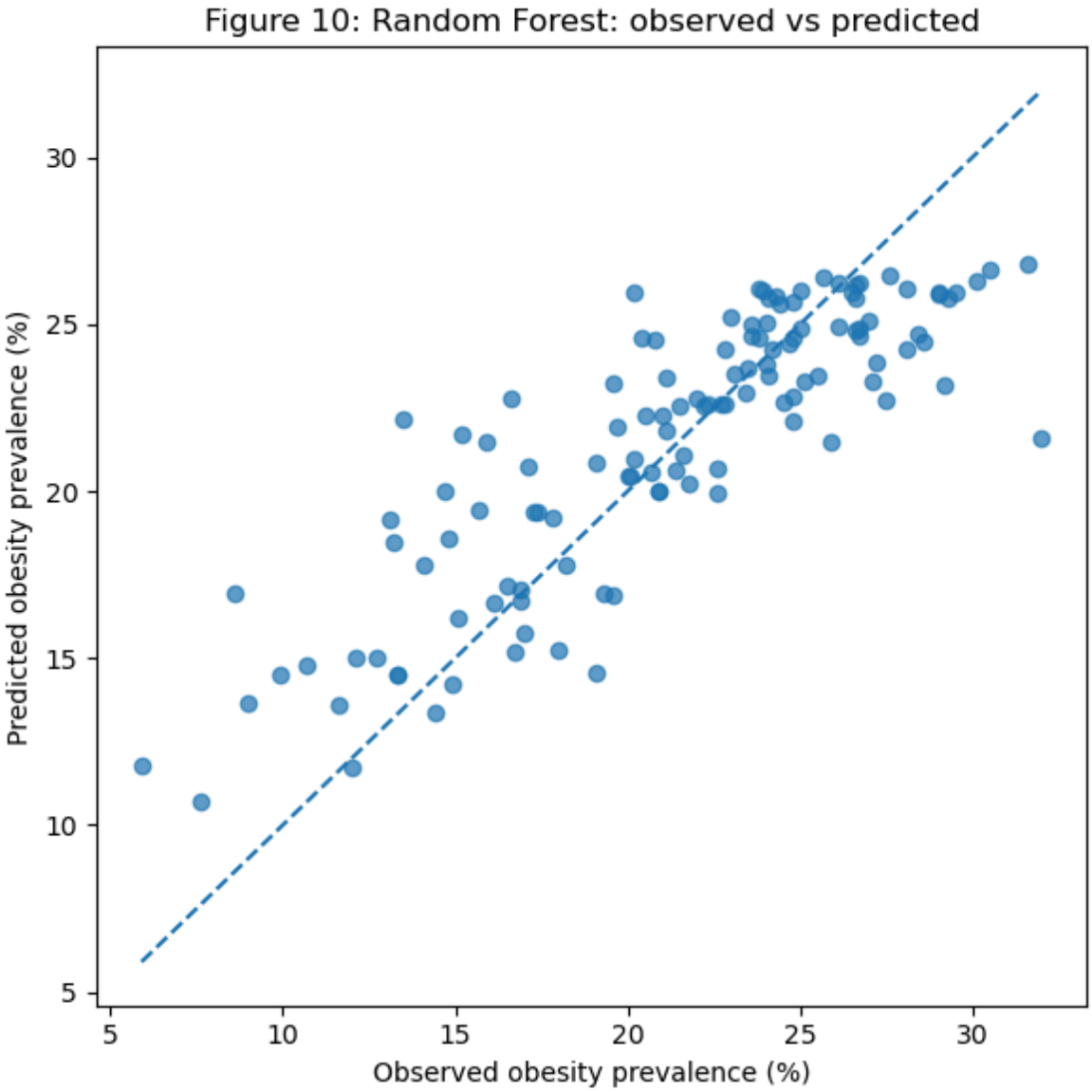


Figure 9: LassoCV: observed vs predicted
Predicted obesity prevalence (%)
Observed obesity prevalence (%)
5
10
15
20
25
30



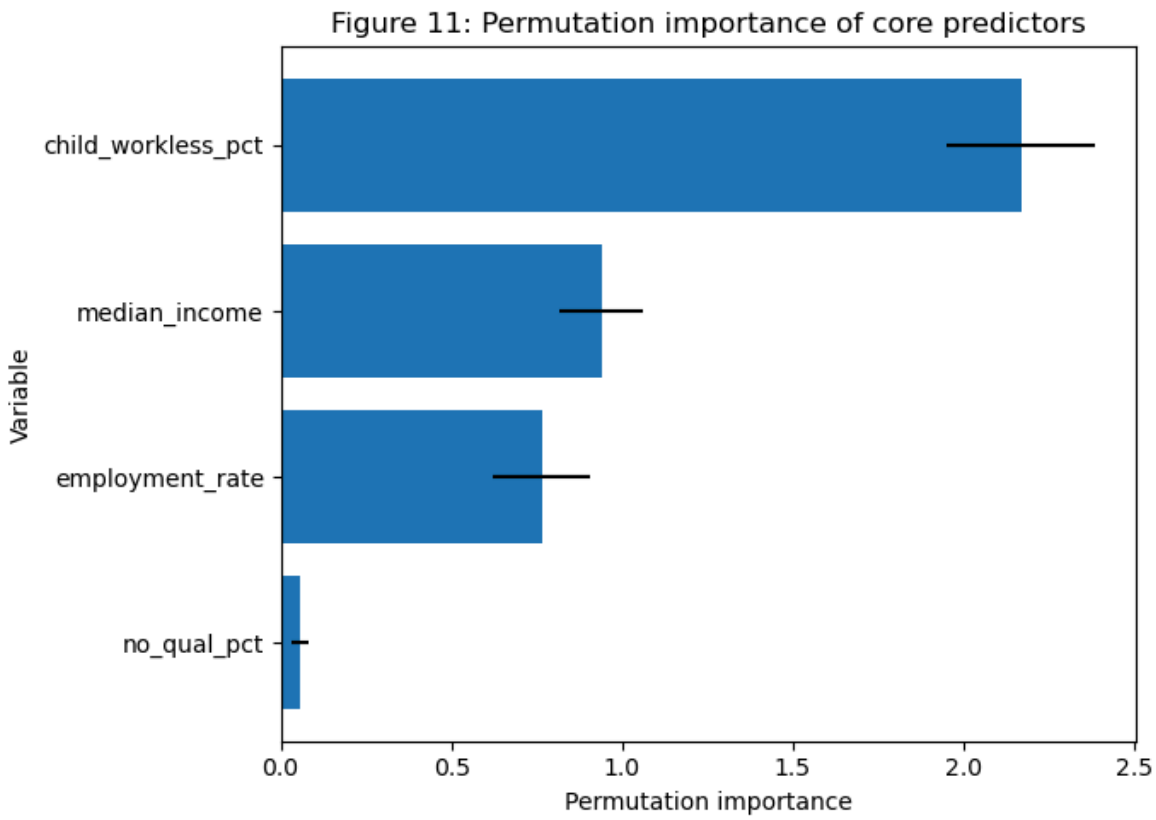
Random Forest outperforms Lasso, with a test set RMSE of 3.036, MAE of 2.294, and R^2 of 0.707, suggesting that the relationship between socio-economic conditions and childhood obesity is not completely linear, and that the tree model is better able to capture more complicated relationships. Figure9-10 show the comparison between the actual and predicted values also shows that Random Forest performs better overall than the linear model.

In [9]:
```
# ========================
# Permutation importance for the RF model
# ========================

import pandas as pd
import matplotlib.pyplot as plt

from sklearn.inspection import permutation_importance

# 1. Calculate permutation importance on the test set
perm_result = permutation_importance(
    estimator=rf_best,
    X=X_test,
    y=y_test,
    n_repeats=30,
```

```
    random_state=42,
    scoring="neg_root_mean_squared_error"
)

# 2. Collation results table
perm_importance_df = pd.DataFrame({
    "variable": predictors,
    "importance_mean": perm_result.importances_mean,
    "importance_std": perm_result.importances_std
}).sort_values("importance_mean", ascending=False)

print("Table 7: Permutation importance (Random Forest, test set):")
display(perm_importance_df.round(4))

# 3. Drawing variable importance diagrams
plt.figure(figsize=(7, 5))
plt.barh(
    perm_importance_df["variable"],
    perm_importance_df["importance_mean"],
    xerr=perm_importance_df["importance_std"]
)
plt.xlabel("Permutation importance")
plt.ylabel("Variable")
plt.title("Figure 11: Permutation importance of core predictors")
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```

Table 7: Permutation importance (Random Forest, test set):

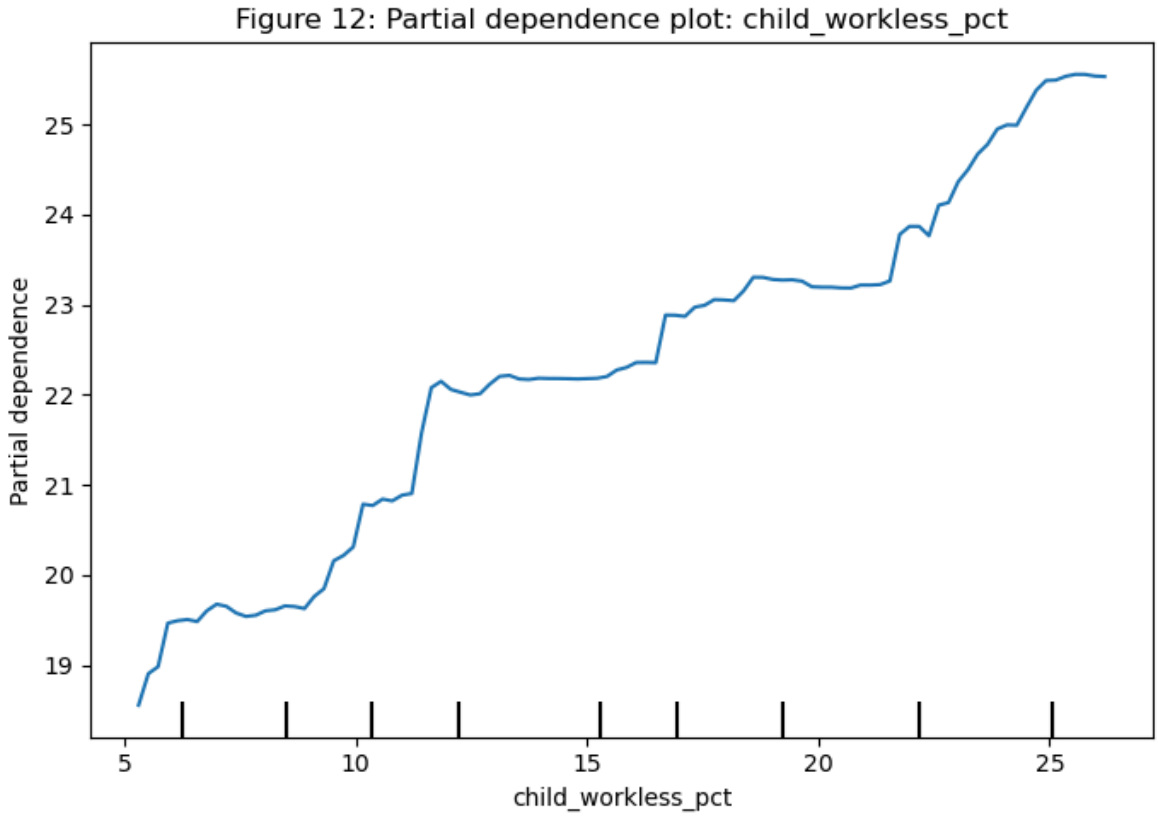
	variable	importance_mean	importance_std
3	child_workless_pct	2.1691	0.2166
0	median_income	0.9402	0.1217
1	employment_rate	0.7645	0.1434
2	no_qual_pct	0.0564	0.0238

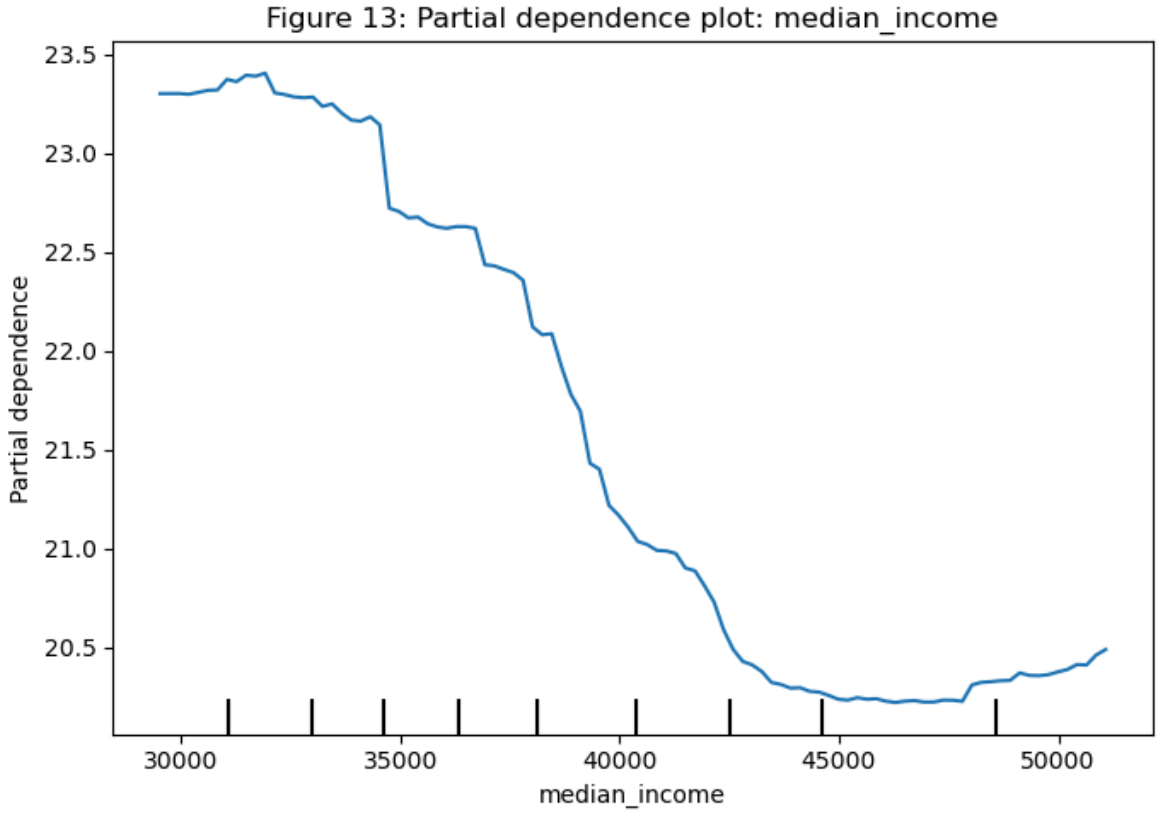


The permutation importance results in Table 7 and Figure 11 show that child_workless_pct is the most important explanatory variable, followed by median_income and employment_rate, while no_qual_pct has a relatively limited contribution.

In [10]:
```
# ========================
# Partial Dependence Plots for the top 2 most important variables
# ========================

import matplotlib.pyplot as plt
from sklearn.inspection import PartialDependenceDisplay

# 1. The top 2 variables with the highest permutation importance were selected.
top2_features = perm_importance_df["variable"].head(2).tolist()
print("Top 2 features for PDP:", top2_features)

# 2. Draw PDP
for i, feature in enumerate(top2_features, 12):
    fig, ax = plt.subplots(figsize=(7, 5))
    PartialDependenceDisplay.from_estimator(
        estimator=rf_best,
        X=X_test,
        features=[feature],
        ax=ax
    )
    ax.set_title(f"Figure {i}: Partial dependence plot: {feature}")
    plt.tight_layout()
    plt.show()
```

```
Top 2 features for PDP: ['child_workless_pct', 'median_income']
```





Partial dependence plots in Figure12-13 also support this conclusion: as child_workless_pct rises, the predicted prevalence of childhood obesity increases overall; as median_income rises, the predicted values decrease overall.This indicates that family working poverty and economic resource level are the two dimensions most closely related to differences in childhood obesity.

Conclusion

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Using London ward as the unit of analysis, this essay examines which local level socio-economic factors are most strongly associated with differences in the prevalence of obesity among sixth grade children. The results illustrated significant differences in the prevalence of childhood obesity across London constituencies, and that these differences were closely related to socioeconomic conditions.The proportion of dependent children in unemployed households, median household income, and employment rates all show strong significance, whether in correlation analyses, Lasso regressions, or Random Forest models.

The model comparison results further show that Random Forest outperforms Lasso, indicating that the relationship between socio-economic factors and childhood obesity is not entirely linear.However, there is an overall consistency in the identification of key variables between the two types of models, which enhances the robustness of the results. As can be seen from the spatial distribution map, the high values of childhood obesity are not randomly distributed, but are more concentrated in areas with relatively weak socioeconomic conditions.

This study also has some limitations. This essay uses ward-level aggregated data, so the conclusions can only explain statistical associations at the district level. Secondly, this paper uses a cross-sectional design, which makes it impossible to make rigorous causal inferences. Nonetheless, this paper suggests that the problem of childhood obesity in London is closely linked to local socio-economic inequalities, and that policies may need to pay more attention to family working poverty and regional resource imbalances if they are to be more effective in reducing levels of childhood obesity.

References

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